

Polymers

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Polymer

The word polymer has a Greek origin. which means many units (parts). Polymer is defined as a chemical substance of a high molecular mass formed by the combination of a large number of simple molecules, called monomers. e.g. Polymer

Polymerisation

The process by which the monomers get combined and transformed into polymers is known as polymerisation.



Difference between Polymers and Macromolecules

Polymers are also called macromolecules due to their large size but converse is not always true. A macromolecule may or may not contain monomer units, e.g., chlorophyll ($\text{C}_{55}\text{H}_{72}\text{O}_5\text{N}_4\text{Mg}$) is a macromolecule but not a polymer since there are no monomer units present so we can conclude that all polymers are macromolecules while all macromolecules may not be polymers in nature.

Classification of Polymers Based on Source of Origin

(i) Natural polymers : The polymers obtained from nature (plants and animals) are called natural polymers.

Examples: Starch, cellulose, natural rubber, proteins, etc.

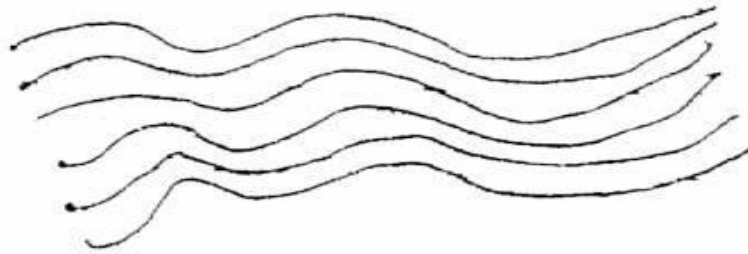
(ii) Synthetic polymers : The polymers which are prepared in the laboratories are called synthetic polymers. These are also called man-made polymers.

Examples: Polyethene, PVC, nylon, Teflon, bakelite, terylene, synthetic rubber, etc.

(iii) Semisynthetic polymers: Polymers obtained by making some modification in natural polymers by artificial means, are known as semi synthetic polymers, e.g., cellulose acetate (rayon), vulcanised rubber etc.

Classification of Polymers Based on Structure

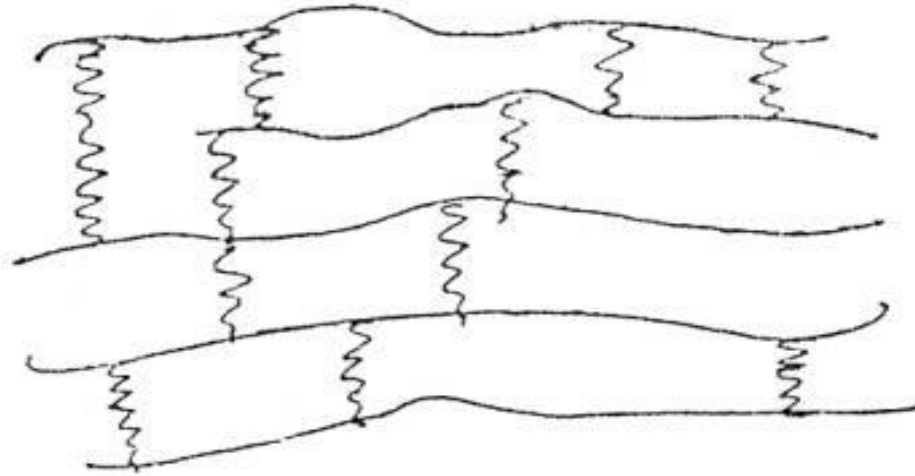
(i) Linear polymers: These are the polymers in which the monomer units are linked to one another to form long linear chains. These linear chains are closely packed in space. The close packing results in high densities, tensile strength and high melting and boiling points. e.g., high density polyethene, nylon and polyesters are linear polymers.



(ii) Branched chain polymers : In such polymers, the monomer units are linked to form long chains with some branched chains of different lengths with source. As a result of branching, these polymers are not closely packed in space. Thus, they have low densities, low tensile strength as well as low melting and boiling points. Some common examples of such polymers are low density polyethene, starch, glycogen etc.



(iii) Cross-linked polymers or network polymers : In such polymers, the monomer units are linked together to form three dimensional network. These are expected to be quite hard, rigid and brittle. Examples of cross linked polymers are bakelite, glyptal, melamine-formaldehyde polymer etc.



Classification of Polymers Based on Mode of Polymerisation

(i) Addition polymers: The polymers formed by the polymerisation of monomers containing double or triple bonds (unsaturated compounds) are called addition polymers. Addition polymers have the same empirical formula as their monomers.

(ii) Condensation polymers : The polymers which are formed by the combination of monomers with the elimination of small molecules such as water, alcohol, hydrogen chloride etc., are known as condensation polymers, e.g., nylon 6,6 is formed by the condensation of hexamethylene diamine with adipic acid.

Difference between Addition & Condensation polymerization

Addition Polymerization	Condensation polymerization (Polycondensation)
1. Involves unsaturated monomere like ethylene, vinyl chloride, styrene etc.	1. Involves substances with atleast 2 functional groups like ethylene glycol (2-OH groups), adipic acid (2-COOH groups)
2. Fast addition of monomers.	2. Step-wise slow addition.
3. At any instant, only monomer and polymers are present.	3. No monomer. Mixtures of dimers, trimers and tetramers are present.
4. Initiator is necessary to catalyse the polymerization	4. Catalyst is not necessary
5. No small molecules are eliminated.	5. Small molecules like H_2O , HCl , CO_2 , CH_3OH are often eliminated.
6. Example	6. Example

Classification of Polymers Based on heat effect

1. Thermoplastics: These are linear polymers with very few cross linkages or no cross linkages at all. The polymeric chains are held by weak VANDER WAAL forces and slide over one another. Due to lack of cross linkages these polymers soften on heating and harden or become rigid on cooling. Thus they can be moulded to any shape. *A thermoplastic polymer is one which softens on heating and becomes rigid on cooling.* Polythene, PVC, polystyrene are addition type thermoplastics and Terylene, nylon are condensation type thermoplastics.

2. Thermosetting plastics : These are normally semifluid substances with low molecular masses. When heated, they become hard and infusible due to the cross-linking between the polymer chains. As a result, they also become three dimensional in nature. *A thermosetting polymer is one which becomes hard on heating .It cannot be softened by heating.* A few common thermosetting polymers are bakelite, melamine-formaldehyde resin and urea formaldehyde resin.

Distinction between Thermoplastic and Thermosetting polymers

Thermoplastic polymers	Thermosetting polymers
1. Linear Polymers.	1. Cross-linked polymers.
2. Weak van der Waals forces and thus soften/melt on heating.	2. Chemical cross-linking make infusible materials.Do not melt on heating.
3. Molten polymer can be moulded in desired shape. It can be remoulded by moulding where they heating again.	3. Cross-linking is usually developed at the time of moulding where they harden irreversibly
4. Examples are polystyrene,PVC,SBR,Teflon, PMMA ,terylene.	4. Examples are glyptol, epoxy polymers,formaldehyde resins.

Classification of Polymers Based on form and use

1. Elastomers :These are rubber like solid polymers in which the polymer chains are held together by weakest intermolecular forces, e.g., natural rubber, buna-S, buna-N etc . The weak binding forces permit the polymers to be stretched. A few ‘cross links’ are introduced in between the chains, which help the polymer to retract to its original position after the force is released as in vulcanised rubber.

2. Fibres :Fibres belong to a class of polymers which are thread-like and can be woven into fabrics. These are widely used for making clothes, nets, ropes, gauzes, etc. Fibres possess high tensile strength because the chains possess strong intermolecular forces such as hydrogen bonding. The fibres are crystalline in nature and have sharp melting.

Stereo specific Polymer

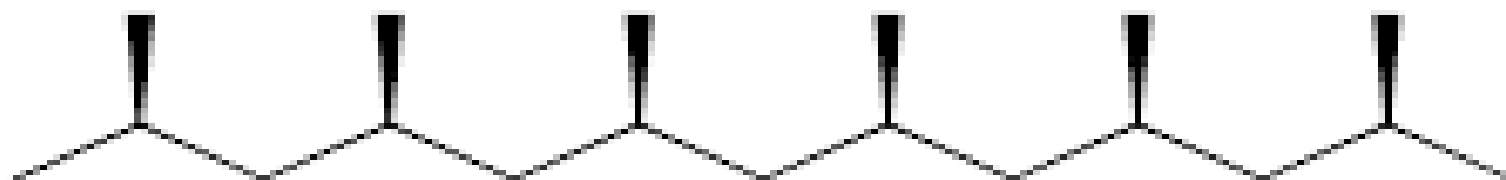
Isotactic: An isotactic polymer is a polymer which has the substituents on the same side of the carbon chain. That means; all the substituents of the polymer material are located on the same side of the backbone of the polymer.

For example, industrially prepared polypropylene is isotactic. Its production method is Ziegler-Natta catalysis. Usually, these polymers are semi-crystalline. They show a helix configuration. Also, this material contains 100% meso diads.

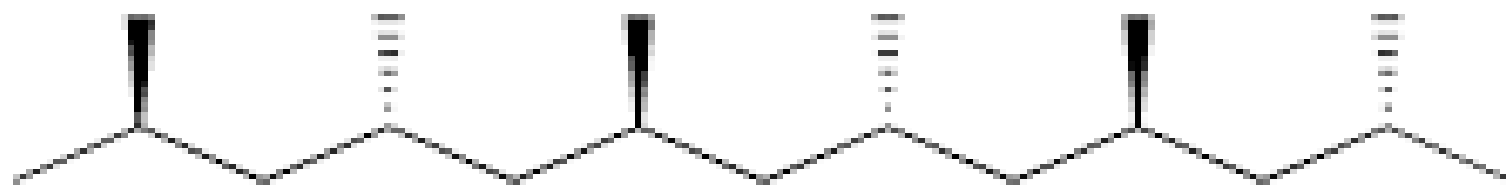
Syndiotactic: Syndiotactic polymers are polymer materials which have the substituents in an alternating pattern. Therefore, substituent groups have alternate positions along the backbone of the polymer.

If we produce polystyrene via metallocene catalysis polymerization, it gives a syndiotactic polystyrene material and it is a crystalline material. The polymer contains 100% racemo diads (the diad contains two units oriented in opposition).

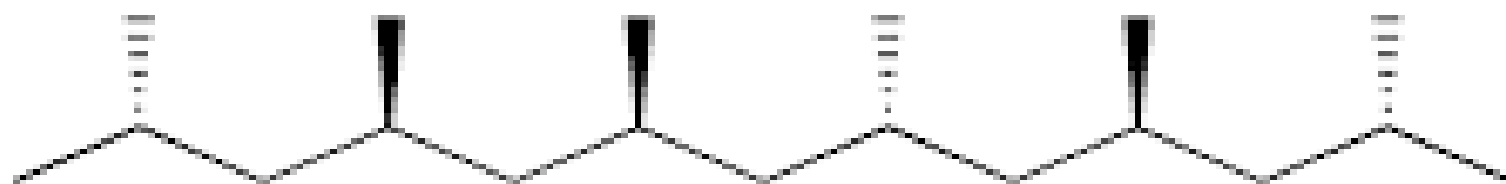
Atactic: An atactic polymer is a polymer material where the substituents in a carbon chain are arranged in a random manner. Usually, polymers that form via free radical polymerization has this structure; for example, polyvinyl chloride. Atactic polymers have an amorphous structure due to the random arrangement of substituent groups.



isotactic



syndiotactic



atactic

Homo-polymer and copolymer

Depending on the monomer molecules polymer may again be classified into two other(different) classes.

Homopolymers :The polymers which are obtained by the polymerisation of a single type of monomer are called homopolymers.

Example:Polyethylene,polystyrene,PVA,polypropylene etc.

Copolymers :The polymers which are obtained by the polymerisation of two or more different monomers are called copolymers.

Example: For example, Buna-S rubber which is formed from 1, 3-butadiene ($\text{CH}_2 = \text{CH} - \text{CH} = \text{CH}_2$) and styrene ($\text{C}_6\text{H}_5\text{CH} = \text{CH}_2$) is an example of copolymer

Polymer Degradation

- Polymer degradation is a change in the properties like tensile strength, color, shape, etc.
- Polymer-based product under the influence of one or more environmental factors such as heat, light or chemicals such as acids, alkalis and some salts.

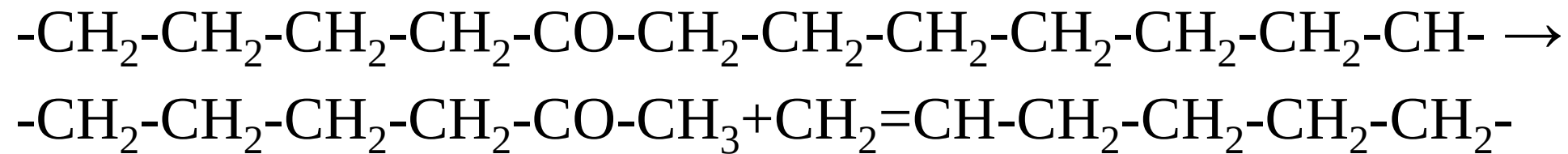
Main factors: solar radiation, temperature, moisture, oxygen. These changes are usually undesirable, such as cracking and chemical disintegration of products.

Solar radiation:

- ☐ Physical changes resulting from exposure to the environment are initiated by chemical bond breaking reactions caused by the absorbed light.
- ☐ The ultraviolet portion of solar energy, with the shortest wavelengths often having the greatest effect.
- ☐ Solar absorptivity is closely related to color, thus samples of different colors will reach different on-exposure temperatures.

Photoinduced degradation:

Most polymers can be degraded by photolysis to give lower molecular weight molecules. Electromagnetic waves with the energy of visible light or higher such as UV light rays and gamma rays are usually involved in such reactions photo degradation of poly alpha methyl,styrene.

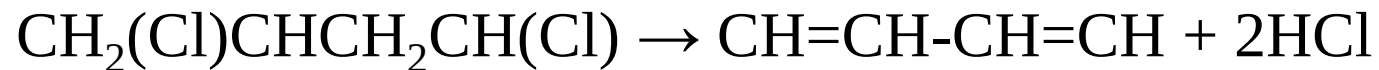


Thermal degradation:

One of the most common forms of degradation is thermal degradation and this takes place in the processing stage of polymers. Generally, polymers are processed at a temperature above its melting point so as to maintain the flow of the melt even after going out of the hot extruder. Unfortunately, not all the polymers are thermally stable above melt temperature.

For instance, PVC is a heat sensitive material that is usually processed above 160°C, but the polymer starts releasing HCl when it passes 120°C. Thus, such polymers are processed with the help of thermal stabilizers.

For example the PVC eliminates HCl, under 100–120 °C.



Mechanical Degradation

Mechanical degradation is also a very common type of degradation that takes place in the processing stage of polymers. When polymers are melted and sheared in the extruder, the shear force breaks the polymer chains thereby reducing the molecular weight of the polymer. Although it is a type of degradation, it aids processing of polymers.

For example, rubber mastication is a process of reduction of molecular weight of raw rubber by passing through pressurized rollers and making it further processable.

UV Degradation

UV degradation takes place in products those are kept on the outside environment. UV radiation from sunlight acts as the energy from the external source to break the bond between the C-C and thereby degrades the polymer. This type of degradation by light is known as Photo-degradation.

UV degradation can yield free radicals that can react with atmospheric O_2 to propagate the degradation reaction that is more of accelerated degradation. This type of degradation is called “Photo-oxidation”.

To avoid such type of degradation by light photo-stabilizers like carbon black is added. This is one of the reasons overhead tanks are mostly black in color.

Oxidative Degradation & Ozone Cracking

Oxidative degradation / Ozone cracking is one of the most common types of degradation with rubbers due to unsaturations in polymer chains of rubber molecule. These unsaturation sites act as the active point for oxidation/ozonation. As mentioned above in photo-oxidation, oxidative degradation is also an accelerative route of degradation.

In order to avoid oxidative degradation/ozone cracking, polymers are incorporated with anti-oxidants & antiozonants. In some cases, rubbers are wax coated to protect them from the attack of atmospheric oxygen and ozone.

Hydrolytic degradation & Biodegradation

Aliphatic polymers with the polyester group, the anhydride group, the amine group in the backbone are susceptible to hydrolysis. Hydrolysis of the polymeric chain leads to low molecular weight stable products. When the molecular weight of the polymer goes very less (say $\sim 5000 - 7000$) they become very small that they can be assimilated by microorganisms.

Assimilation of low molecular weight products by microorganisms is known as biodegradation. Biodegradation leads to mineralization. Mineralization is the process of conversion of organic carbon to inorganic CO_2 and H_2O .

Polymers like PLA, PBS are biodegradable as initially they are hydrolyzed followed by assimilation by microorganisms. Thereby they don't stay in the atmosphere for centuries like conventional polymers.

Moisture:

- Moisture, in combination with solar radiation, contributes significantly to the weathering of many polymeric materials
- Mechanical stresses imposed when moisture is absorbed or desorbed and to the chemical participation of moisture in the chemical evolution cause weathering
- The span of time over which the precipitation occurs and the frequency of wetness are important in the weathering of materials
- Water absorption in the surface layers produces a volume expansion which places mechanical stress on the dry subsurface layers.
- Drying out of the surface layers would lead to a volume contraction.
- The hydrated inner layers resist this contraction, leading to surface stress cracking.

Types of degradation

Polymer degradation is mainly two types

1.Chain end degradation: The successive release of monomeric units from the end of the polymer chain because of physical or chemical causes can be termed as chain end degradation. This degradation starts from the chain ends which is actually reverse of the propagation step of the polymerization. That's why chain end degradation is also termed as depolymerisation.

In chain end degradation molecular weight of the polymer decreases slowly and a large quantity of the monomer is liberated simultaneously

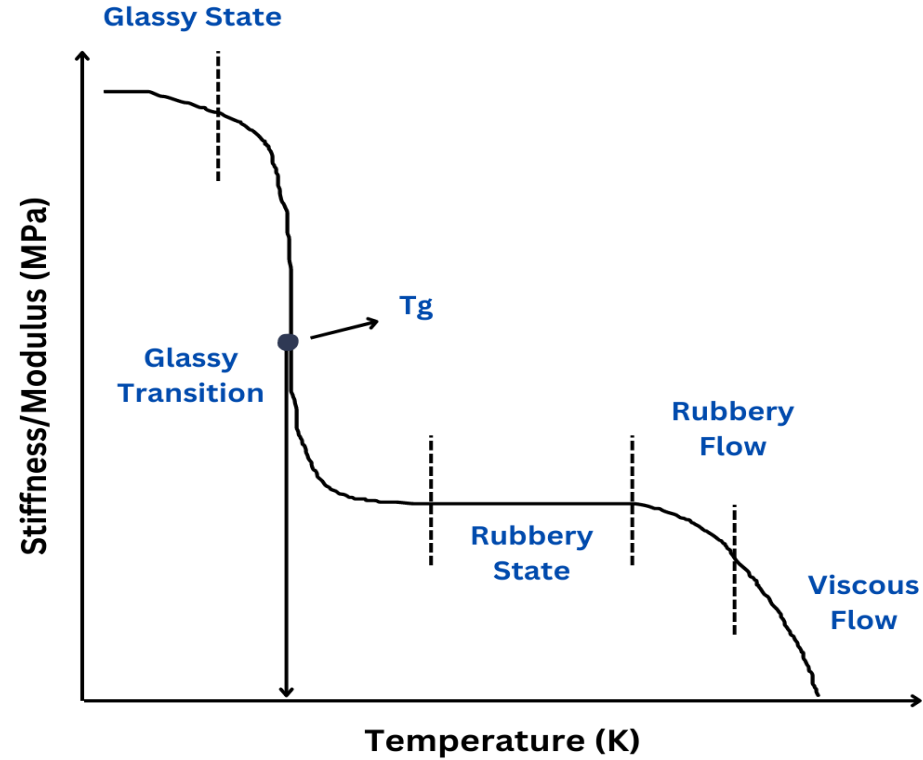
Random Degradation: Random degradation is the polymer degradation where degradation may occur at any random point along the polymer chain instead of the chain ends. Random degradation is the reverse of the poly condensation process.

Here polymer degrades to lower molecular weight fragments but (unlike in chain end degradation) practically no monomer is liberated.

Almost all polymer can undergo random degradation.

Glass transition temperature(T_g).

When plastic or rubber is cooled up to a certain temperature, it becomes so hard and brittle that it breaks into pieces on the application of stress. So, the temperature below which the polymer becomes hard, brittle, and glassy and above which it is softer and flexible is known as **glass transition temperature (T_g)**.



The glass transition is a property of only the amorphous region of a semi-crystalline solid whereas the crystalline portion remains crystalline during the glass transition.

It is important to note that the transition does not occur suddenly at a unique temperature but rather over a range of temperatures. The temperature in the middle of the inclined region is taken as the T_g .

Significance of glass transition temperature (T_g)

- Glass transition temperature (T_g) is used as a measure for evaluating the flexibility of a polymer and the type of response the polymeric material would exhibit to mechanical stress.
- Glass transition temperature (T_g) tells us that, above it, a polymer will be soft and flexible and exhibit viscoelasticity (Delayed elastic response) and below it, a polymer will be hard and brittle and will possess dimension stability.
- The glass transition temperature (T_g) value along with the T_m value indicates the temperature region at which a polymeric material transforms from a rigid state to a soft viscous state. This helps in choosing the right processing temperature which is the temperature region in which the material can be converted into the finished products through different processing techniques such as molding calendaring and extrusion.

Factors affecting glass transition temperature (T_g)

1. Chain flexibility and rigidity

Stiffening groups in the polymer chain reduce the flexibility of the chain and increase the value of glass transition temperature (T_g). In other words, we can say that the greater the intrinsic chain flexibility, the smaller will be the value of glass transition temperature (T_g).

2. Intermolecular Forces

Stronger intermolecular forces due to dipole forces, H-bonding, etc decreases the mobility of the chain which increases the value of T_g . For example, T_g of PVC is greater than T_g of polypropylene because PVC has stronger intermolecular forces than polypropylene because of the dipole-dipole forces from the C-Cl bond.

3. Molecular weight

Glass transition temperature (T_g) is directly proportional to the molecular weight of the polymer that is when the molecular weight of the polymer increases, T_g also increases.

4. Pendant Groups

Pendant groups have a considerable amount of effect on chain mobility. Bulky pendant groups like benzene can catch on neighboring chains like a "fish hook" and restrict rotational freedom which increases the glass transition temperature (T_g).

For example, a Big group like the adamantyl group will not only act as a "fish hook" but also it will act as a "downright boat anchor" that will increase the T_g .

T_g of poly(ether ketone) with an adamantyl group is $225\text{ }^{\circ}\text{C}$ which is greater than the T_g of poly(ether ketone) that does not have an adamantyl group.

5. Cross-linking

The greater the degree of cross-linking, the higher will be the T_g because the presence of cross-links between chains restricts rotational motion.

6. Plasticizers

Plasticizers are low molecular weight compounds that are added to plastics to increase their flexibility and workability. This weakens the intermolecular forces between the polymer chains and increases the mobility thus decreasing the glass transition temperature (T_g).

7. Copolymerization

Random copolymers have lower T_g since it tends to promote the disorder, reduce molecular packing, and reduce the inter-chain forces of attraction.

Some Important Polymers

Polythene:

(a) Low-density polythene (LDP) or Low-density polyethylene (LDPE)

LDPE is a very flexible and tough polyethylene prepared by heating ethylene to 350-570 K under a pressure of 1000-2000 atm in the presence of a free radical or peroxide.

LDPE which is produced has a molecular structure due to hydrogen abstraction.

Preparation of LDPE with the mechanism of free-radical polymerization.

LDPE is prepared by free-radical polymerization. In this process, free radicals not only add to the double bond of ethylene but also abstract hydrogen from a chain already formed. This abstraction generates a new free radical on the chain, which a chain can grow.

(LDPE)

heating ethylene to 350-570 K under a pressure of 1000-2000 atm in the presence of a free radical or peroxide.

LDPE has a highly branched

which the growing free radical abstracts hydrogen from a chain already formed. This abstraction generates a new free radical on the chain, which a chain can grow.

Properties of LDPE

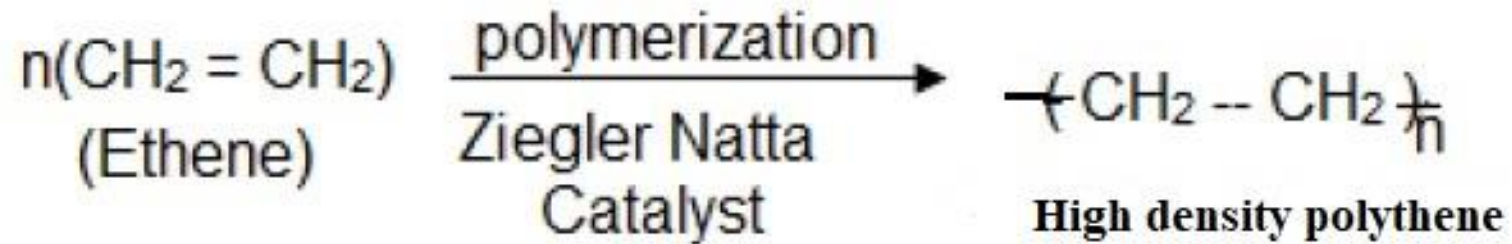
1. It is a transparent polymer of moderate tensile strength and high toughness.
2. It is chemically inert and slightly flexible.
3. It is a poor conductor of electricity.
4. It has a low heat deflection temperature and melting point as compared to HDPE.
5. It has a high degree of branching which results in a lower degree of crystallinity in the range of 40-55 %.

Uses of LDPE

1. It is widely used as a packaging material in the form of thin plastic films, bags, etc.
2. It is used as an insulator for electrical wires and cables.
3. It is used in the manufacturing of squeeze bottles, toys, and flexible pipes.

Preparation of HDPE

HDPE is prepared by coordination polymerization of ethene. In this process, ethene (in a hydrocarbon solvent) is heated to 333-343 K under a pressure of 6-7 atmosphere in a presence of a Ziegler-Natta catalyst.



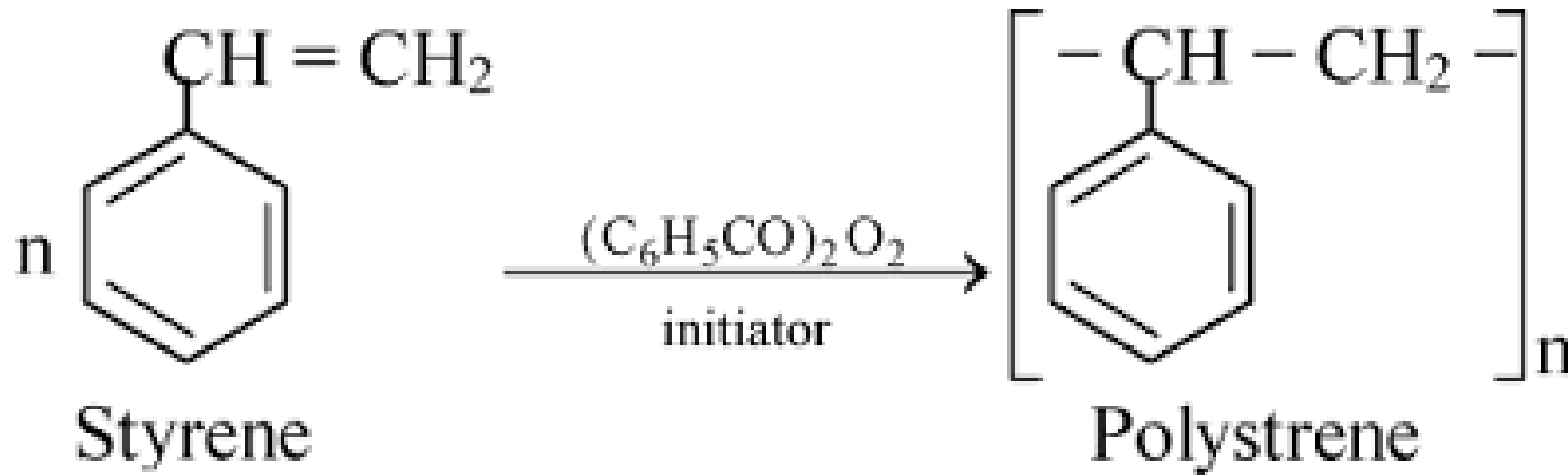
Properties of HDPE

1. It is translucent.
2. It is chemically inert but has greater toughness, hardness, and tensile strength than low-density polythene (LDPE).
3. It has a low degree of branching which results in a higher degree of crystallinity in the range of 70-80 %.
4. It has a high density (0.97 g/cm^3) and high melting point (403 K) as compared to LDPE.

Uses of HDPE

1. It is used in the manufacturing of containers (buckets, tubs, etc) housewares, pipes, bottles, and toys.
2. Due to high strength and UV resistance, it is used in the piping system of homes, cities, and towns.
3. It is used in heavy machinery in factories due to its stress crack resistance property.

2. Polystyrene (Styrene):Polystyrene is prepared by free radical addition polymerization of styrene in the presence of benzoyl peroxide as a catalyst.



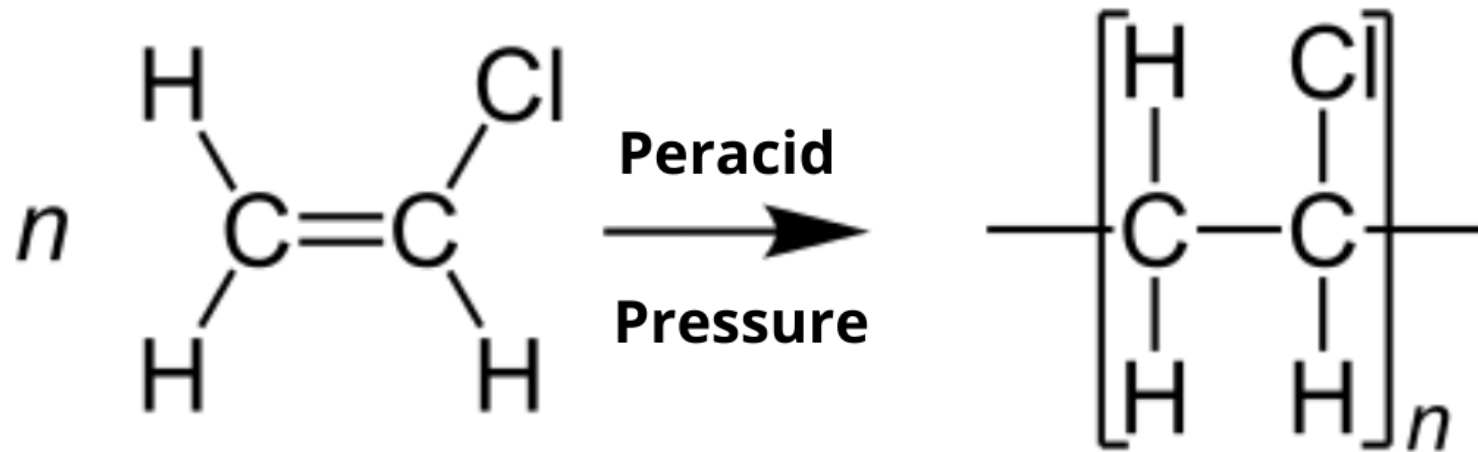
Properties of Polystyrene

1. Polystyrene exists in an amorphous state due to the presence of bulky phenyl groups which makes packing of Polystyrene inefficient.
2. Polystyrene is non-polar having a melting point of 240 °C.
3. Polystyrene is a transparent polymer having a good optical property that allows high transmission of all wavelengths.
4. Polystyrene is hard but brittle due to the chain-shifting effect of the benzene ring.
5. Polystyrene has good insulation properties. That's why it is used in making Polystyrene insulation products.

Uses of Polystyrene

1. Polystyrene is used in making 'throw away' drinking cups, soft drinks, and baby feeding bottles.
2. By adding color and filler, it can be used for making toys and household items.
3. When gas is blown into Polystyrene liquid, it foams and hardens to form styrofoam which is used for making ice chests and disposable coffee cups.
4. Polystyrene is also used as packing material for shipping instruments and appliances, and it is widely used for home insulation.
5. Medically it is used for sterilizing test tubes, diagnostic components, and other medical devices

3. Polyvinylchloride (PVC): The monomer vinyl chloride is treated with peracid under pressure to obtain Polyvinyl Chloride (PVC)



Properties of Polyvinyl Chloride (PVC)

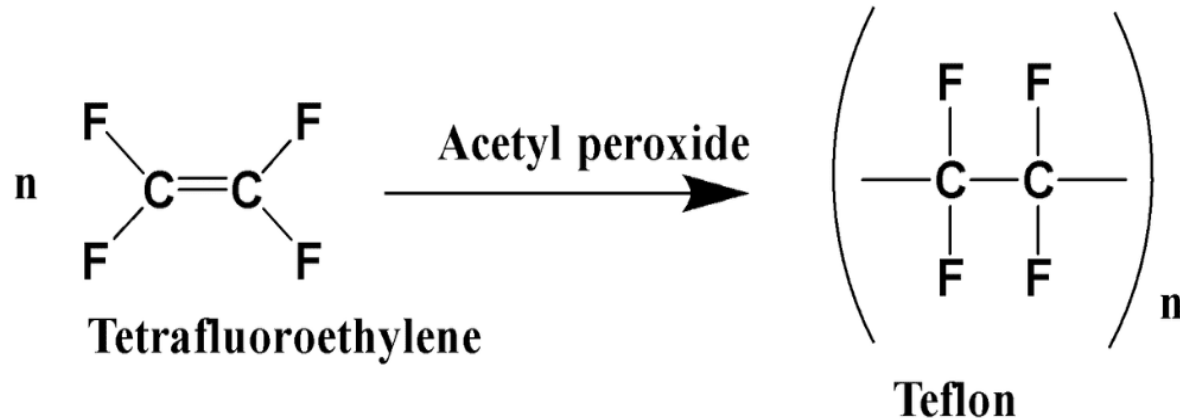
1. It is a linear polymer and thermoplastic in nature.
2. It is a white brittle solid, hard, rigid material which tends to stick to the metallic surface.
3. It is insoluble in all hydrocarbon solvents.
4. It has a melting point of 212°C and glass transition temperature (T_g) is 80°C .

Uses of Polyvinyl Chloride (PVC)

1. It is used for making raincoats, handbags, shower curtains, vinyl flooring, water pipes, etc.
2. Around the world, over 50 % of PVC manufactured is used in construction. As a building material, it is cheap, durable, and very easy to assemble. It is a rigid material and is used to make pipes, panels, and molded objects.
3. PVC is commonly used as insulation on electric wires.
4. PVC is used in the healthcare sector for the manufacturing of blood collection, ostomy, and urine collection products.

5. Teflon:

Teflon is prepared by heating tetrafluoroethylene in the presence of peroxides or ammonium persulphate catalyst at high pressures



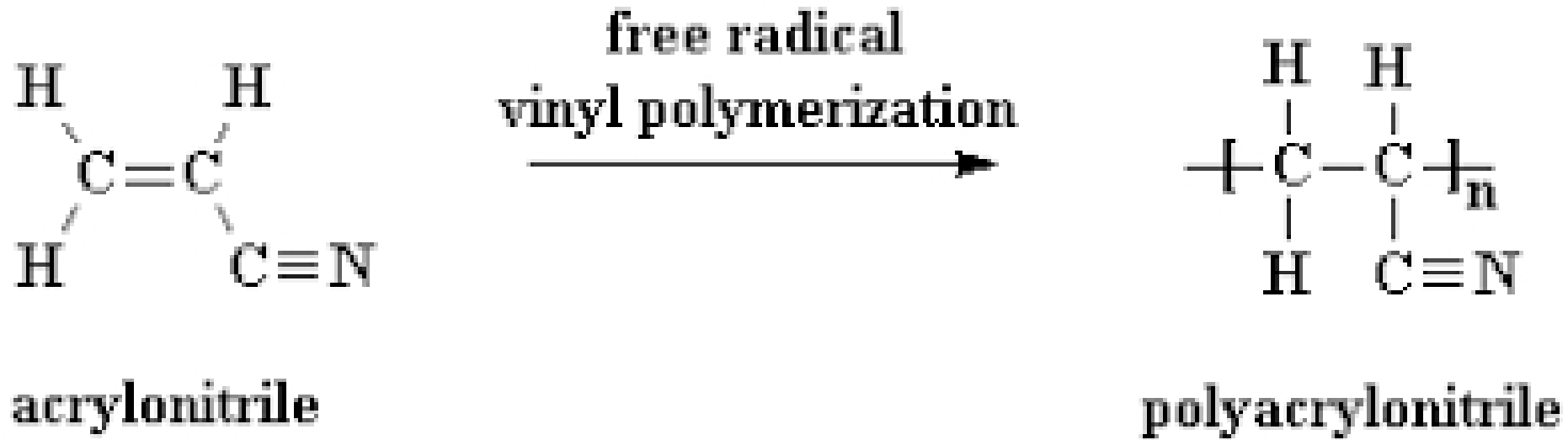
Properties of Teflon

1. Teflon is flexible and inert to solvents and boiling acids even to aqua regia and is stable up to 598K temperature.
2. It has a low coefficient of friction and low dielectric constant due to the highly symmetric structure of the macromolecules.
3. It has good insulating power in hot and wet environments.
4. Teflon maintains high strength, toughness, and self-lubrication at low temperatures of $-268.15\text{ }^{\circ}\text{C}$.

Uses of Teflon

1. Teflon is used for making non-stick utensils due to its great chemical inertness and high thermal stability.
2. It is also used for making gaskets, pump packings, valves, seals, non-lubricated bearings, filter cloth, etc.
3. It is also used as electrical insulations in connector assemblies, cables, and in printed circuit boards.
4. Teflon film is also widely used in the production of carbon fiber composites as well as fiberglass composites.

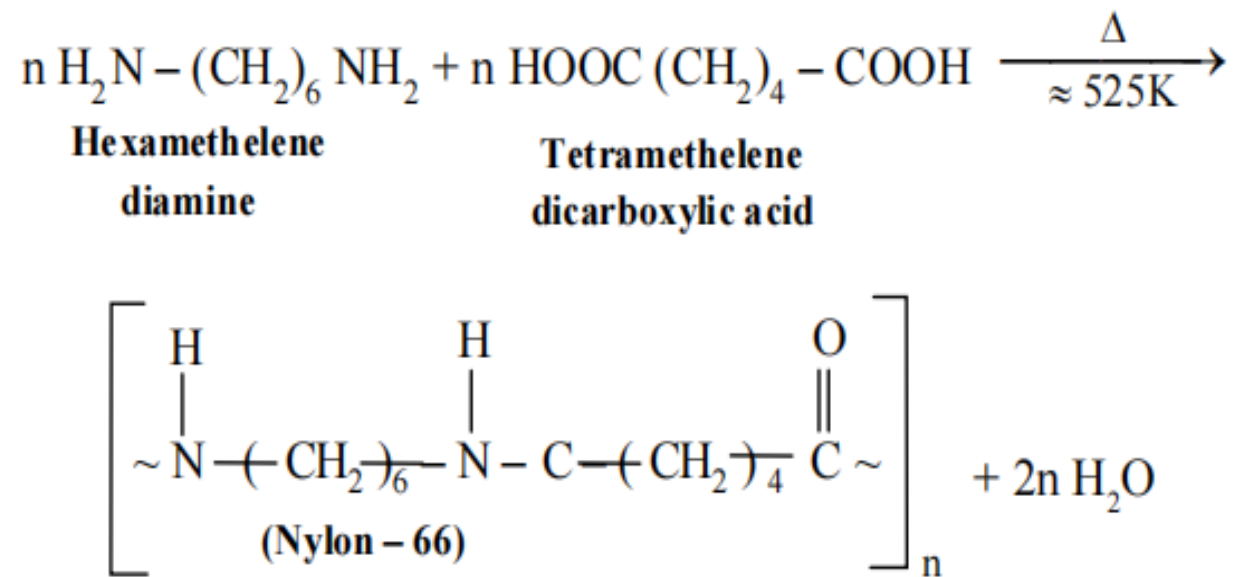
6. Polyacrylonitrile



It is used as a substitute for wool in making commercial fibres as orlon or acrilan.

Nylon-66

It is prepared by the condensation polymerization of adipic acid and hexamethylenediamine. The acid and the amine first react to form a salt which when heated to 525 K under pressure, undergoes polymerization with the elimination of water as steam and the nylon is produced in the molten state.



The polyamides are identified by numbers. These numbers refer to the number of carbon atoms in diamine and in the dibasic acid. As in the above case, the carbon atoms are 6 in each case, therefore the product is described as _nylon-66.

Properties of Nylon 6, 6

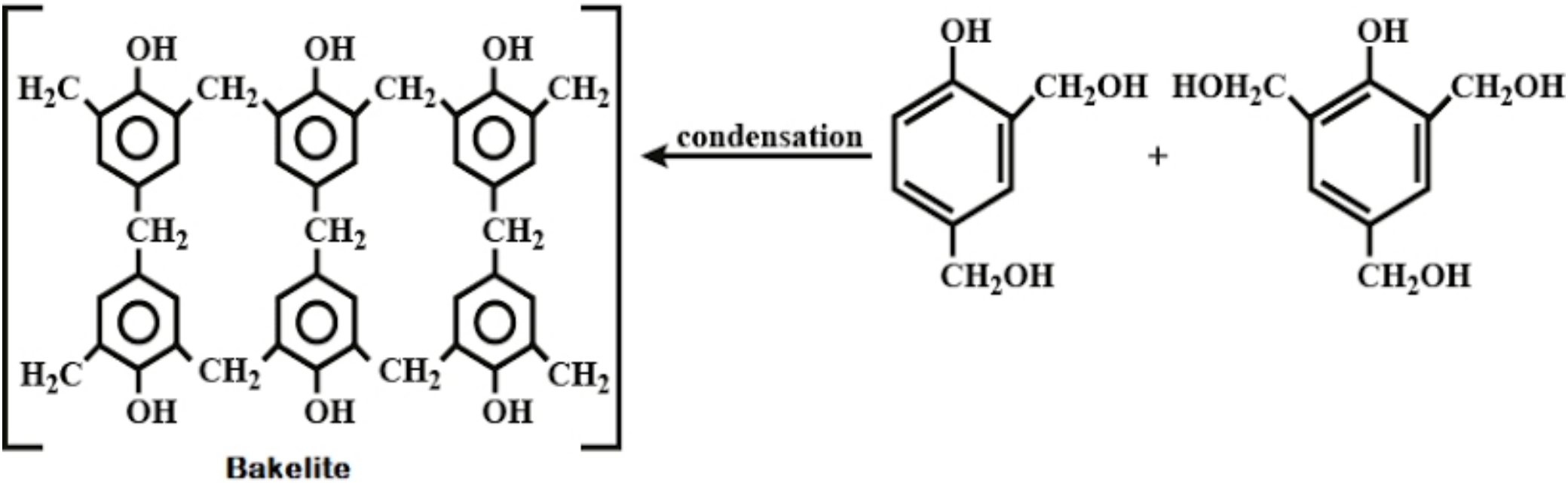
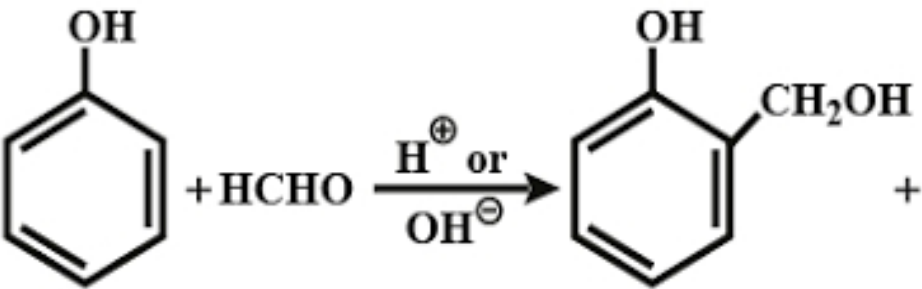
1. It has a high melting point and high heat deflection temperature.
2. It offers better resistance to acids and has high stain cleanability.
3. It is more difficult to color due to dense structure with small and evenly spaced pores.
4. It exhibits greater mold shrinkage, lower expansion, and improved thermal and electrical conductivity.

Uses of Nylon 6,6

1. It is used for making swimwear, hoses, and machine parts due to its waterproof property.
2. It is also used for making glass-reinforced laminates, fasteners, and conveyor belts.
3. It is used in fibers for textiles, carpets, and molded parts.
4. It is also used for making surgical sutures and strings for musical instruments.

Phenol-Formaldehyde Polymer

These polymers are obtained by the condensation reaction of phenol with formaldehyde in the presence of either acid or a base catalyst. The reaction involves the formation of methylene bridge at ortho, para or both ortho and para positions. A linear or cross linked material is obtained depending upon the condition of reaction.



Properties of Bakelite

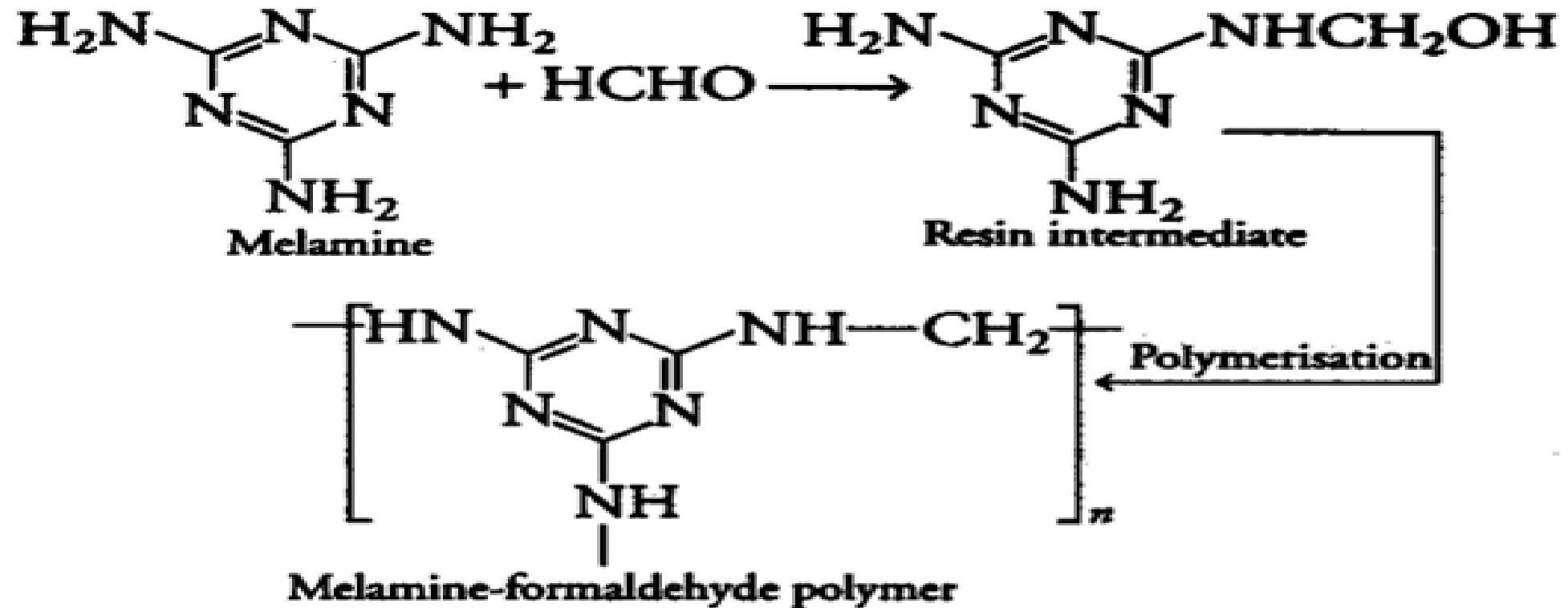
1. It can be quickly molded and very smooth molding can be obtained from this polymer.
2. Bakelite moldings are heat-resistant and scratch-resistant. They are also resistant to several destructive solvents.
3. Bakelite is resistant to electric current due to its low electrical conductivity.
4. Bakelite comes in both a low degree of polymerization which is soft and a high degree of polymerization which is hard and rigid.

Applications of Bakelite

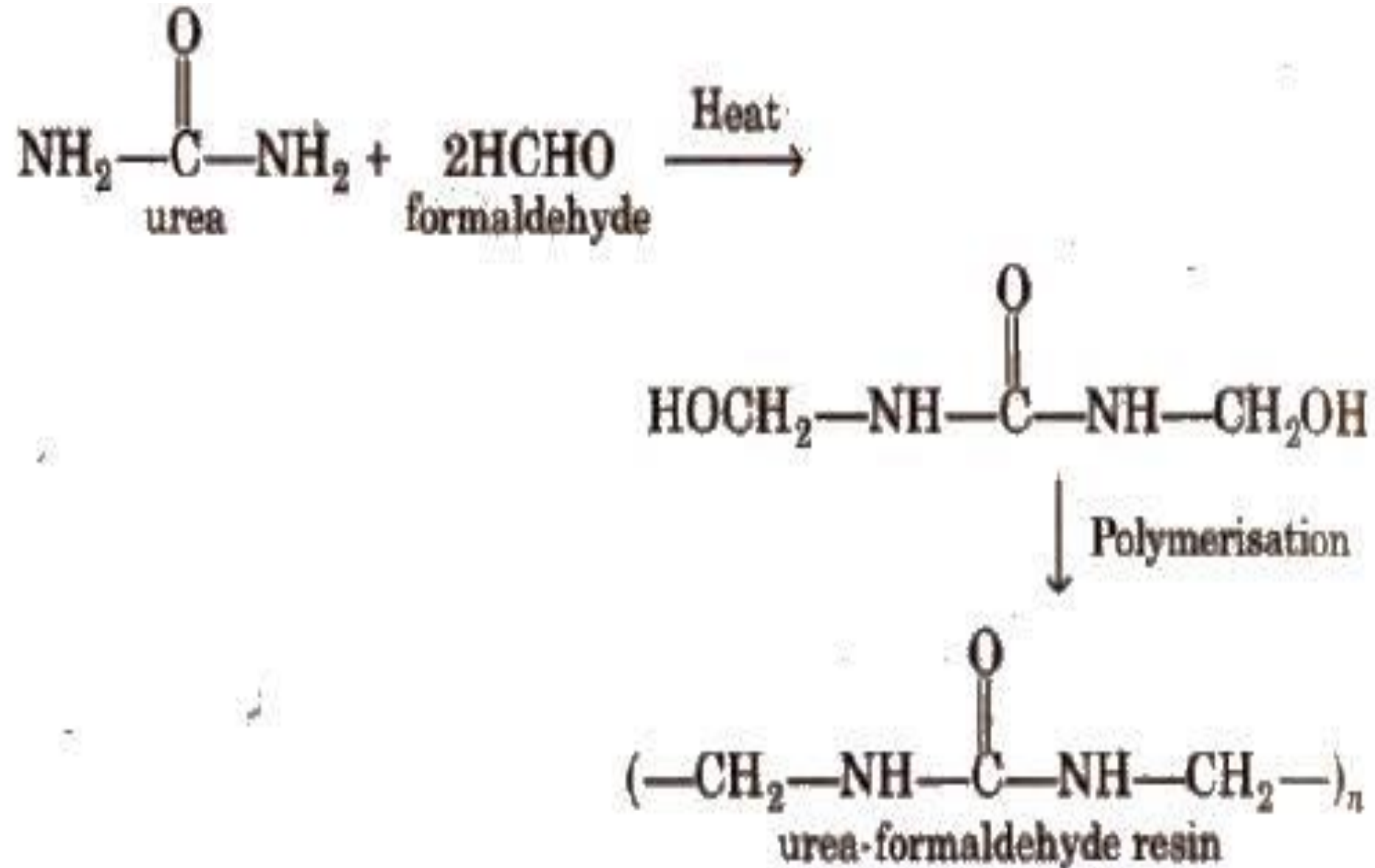
1. Soft Bakelite with a low degree of polymerization is used as binding glue for laminated wooden planks and in varnishes and lacquers.
2. It is used for the manufacture of combs, Formica table-tops, fountain pen barrels, phonograph records, computer discs, etc.
3. It is widely used in making electrical goods like switches, plugs, handles, or various utensils due to its electrical insulating property.
4. Sulphonated Bakelites are used as ion exchange resins.

Melamine-formaldehyde Resin

It is a copolymer formed by the polymerisation of melamine (which is a heterocyclic triamine)



Urea-formaldehyde Resin



Properties of Urea formaldehyde Resin

High tensile strength – The maximum stress that a material can withstand while being stretched or pulled before breaking.

High flexural modulus – The ratio of stress to strain that a material can endure while bending before it yields.

High heat-distortion temperatures – At what temperature the material will begin to “soften” when exposed to a fixed load at elevated temperatures.

Low water absorption

Mold shrinkage – When the volume of the molten plastic filled inside the cavity of a mold shrinks during the process of cooling and solidifying.

High surface hardness

Elongation at break – The ratio between changed length and initial length after the material breaks.

Volume resistance – The electrical resistance of a body to a current passing through its bodily substance.

Elastomers

Temperature: The specific working temperature of elastomers vary depending on the factors like media compatibility, seal design, and dynamic and static operation.

Low-temperature flexibility: The rate of recovery of elastomeric material can be studied by subjecting the material to low-temperature retraction.

Hardness: The measurement of the material's resistance towards the deforming force for a defined length of time is done by measuring the hardness. It differs from material to material. The soft compounds deform easily and have high friction, while the harder compounds have high resistance and low friction.

Ageing: This property helps to understand the behaviour of a material when exposed to heat. If the elastomers are pushed beyond their ageing resistance, they will suffer from hardening, cracking, and splitting.

Colour: Colouring is used mainly to differentiate between the compound grades based on their usage.

Elongation at break: This property is used for testing the moment of rupture when the material is under tensile stress..

Following are the two types of elastomers:

- **Saturated Elastomers:** Sulphur vulcanization can not cure them. They showcase superior stability against oxygen, radiation, heat, and ozone. Comparably they are less reactive. Their reactivity is limited to certain circumstances and conditions. Polyacrylic rubber and silicone rubber are examples of saturated elastomers.
- **Unsaturated Elastomers:** They can be cured with the Sulphur vulcanization process. Butyl rubber and natural polyisoprene are examples of unsaturated elastomers

Application of Elastomer

Some of their applications are listed below-

Motor vehicles: Some elastomers like thermosets don't melt easily, making them efficient in building seals, tyres various components throughout the automobile design. Especially those components which will be exposed to heat during the functioning. The material of the type polybutadiene offers extraordinary wear resistance hence they are preferable in building tyres.

Consumer products: This comprises the widest range of products starting from shoe soles to baby pacifiers and many more miscellaneous.

Constructions: Adhesives and sealants materials enfolded under elastomers, which are an unavoidable part of any constructions. Especially for filling the gaps.

Industrial products: Elastomers are hugely used in making industrial tools, appliances, belts, molds, lubricants, etc.

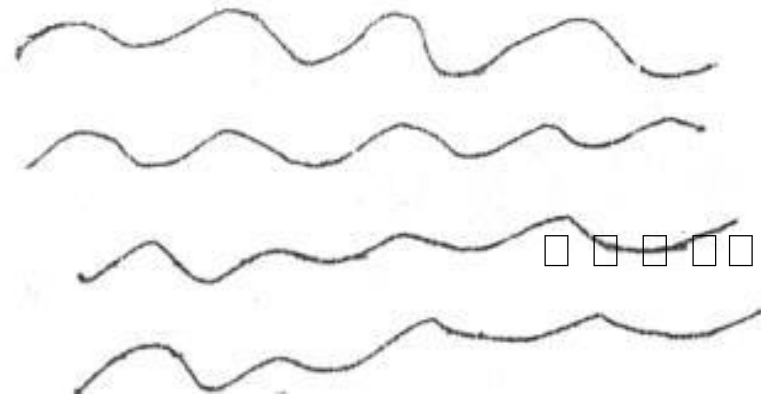
Wire and cable: Material needed to build wires should have high resistance to heat, be easily reshaped(elongated), and provide insulation. Elastomers like neoprene are perfect for this.

Medical products: Medical field needs a wide range of products like prosthetics, lubricants, and moulds with superior class of chemical and thermal resistance. Elastomer like silicon has widely used the material to build them and many other goods.

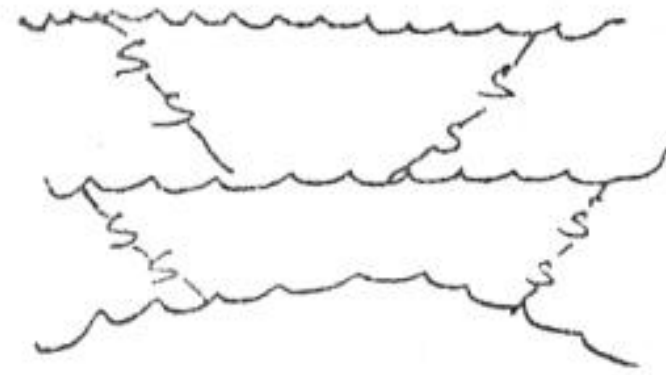
Vulcanization of Rubber

The properties of natural rubber can be modified by introducing -S-S- polysulphide crosslinks in its structure. This process of introducing -S-S- crosslinks in the structure of natural rubber by heating with sulphur at 110°C is called vulcanization of rubber.

Vulcanization is carried out by adding sulphur (3-5%) and zinc oxide to the rubber, and then heating the object at about 110°C for about 20-30 minutes. Zinc oxide accelerates the rate of vulcanisation. Vulcanization introduces polysulphide (-S-S-) bonds between the adjacent chains. These crosslinks tend to limit the motion of chains relative to each other.



Natural rubber



Vulcanized rubber

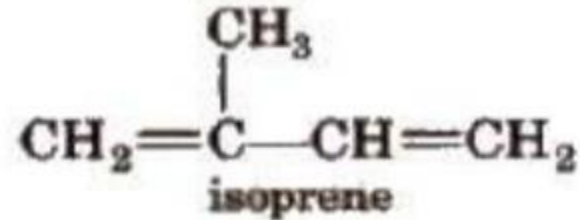
Fig. : Process of vulcanization of rubber

Comparison of some properties of natural rubber and vulcanized rubber

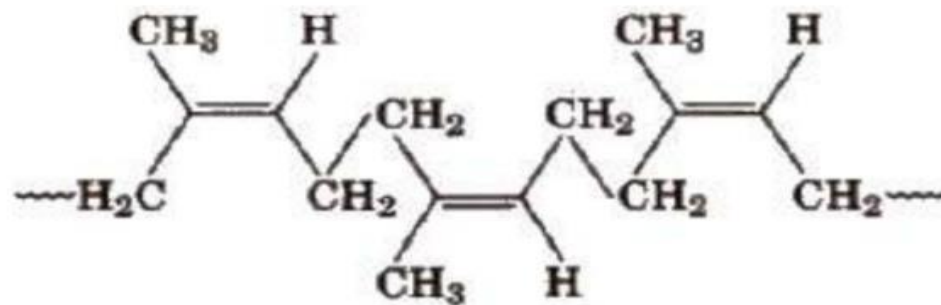
Natural Rubber	Vulcanized rubber
1. Natural rubber is soft and sticky.	1. Vulcanized rubber is hard and non-sticky.
2. It has low tensile strength.	2. It has high tensile strength.
3. It has low elasticity.	3. It has high elasticity
4. It can be used over a narrow range of temperature (from 10^0 to 60^0C)	4. It can be used over a wide range of temperature (from -40^0 to 100^0C).
5. It has low wear and tear resistance.	5. It has high wear and tear resistance.
6. It is soluble in solvents like the ether, carbon tetrachloride, petrol, etc	6. It is insoluble in most of the common solvents.

Natural rubber

Natural rubber is a coiled linear 1, 4-polymer of isoprene



In the polymer chain of natural rubber, the residual double bonds are located between C₂ and C₃ of the isoprene unit. All these double bonds have cis configuration, and thus natural rubber is cis-1,4-polyisoprene.

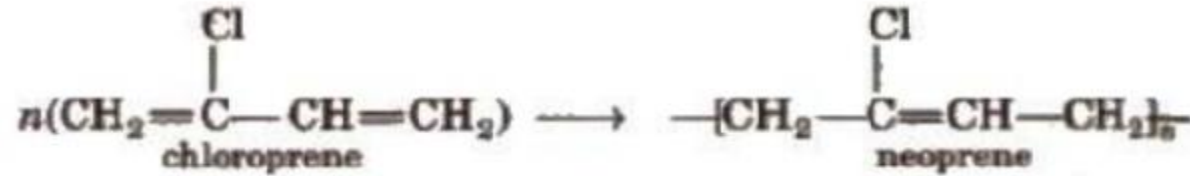


A section of the polymeric chain of natural rubber

Synthetic rubber

Neoprene

Polymer formed by polymerisation of chloroprene is neoprene or synthetic rubber



Properties

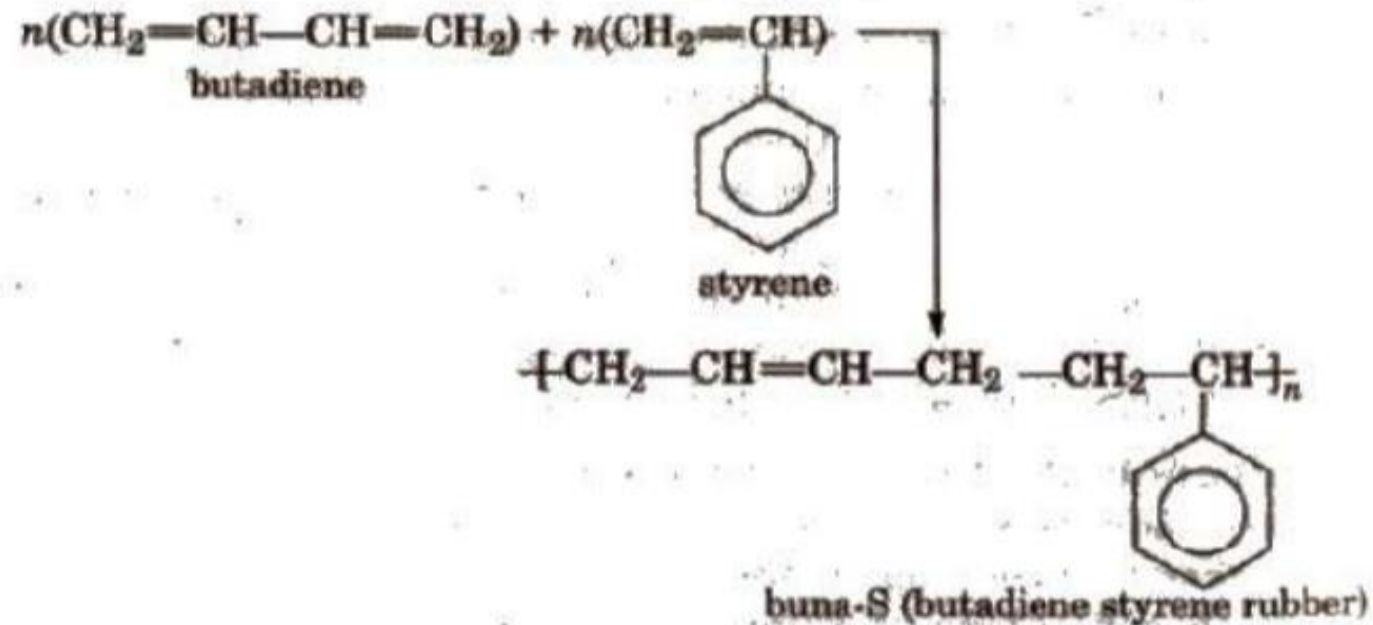
- It has higher resistance to oxidizing agent.
- It is soluble in polar solvent.
- It is not attacked by oil, alkali solution.
- Its tensile strength is less than natural rubber.

Uses of neoprene rubber

- For manufacturing good quality of tier
- It is used for the manufacturing conveyers belts, gasket and hoses.

SBR (Styrene Butadiene Rubber)

- It is used for the manufacture of gaskets used for sealing refrigerator, cabinet, doors etc.
- It is also used for the manufacture of safety glass. butadiene rubber may be represented as:



Properties of SBR

- It is high resistance to abrasion, ozone and weathering action.
- It is readily oxidizes.
- It has high load bearing capacity

Uses of SBR

- Used for manufacturing lighter duty tyres, hoses, belts, un vulcanized sheet, gum, floor mat, shoe's soles, electrical insulators, adhesives, tank lining, coated fabrics etc.

Composite materials

A composite material is a material that consists of two or more physically distinct phases whose combination produces aggregate properties that are different those of its constituents.

Composite materials, or shortened to composites, are microscopic or macroscopic combinations of two or more distinct engineered materials (those with different physical and/or chemical properties) with a recognizable interface between them in the finished product. For structural applications, the definition can be restricted to include those materials that consist of a reinforcing phase such as fibers or particles supported by a binder or matrix phase. Wood composites are commonly seen examples of composite materials

Composition of composites

They generally have two phases

- Matrix phase

It is the continuous material constituent which encloses the composite and give it its bulk form

Matrix phase may be metal ,ceramic or polymer

- Dispersion phase

It is the structure constituent ,which determines the internal structure of composite.Also called reinforcement.

Fiber,particles,flakes etc.

Classification of Composites

Matrix

- Organic Matrix Composites (OMCs)
- Polymer Matrix Composites (PMCs)
- carbon-carbon composites
- Metal Matrix Composites (MMCs)
- Ceramic Matrix Composites (CMCs)

• Reinforcements:

- Fibres reinforced composites
- Laminar composites
- Particulate composites

Carbon Fiber is a polymer and is sometimes known as graphite fiber. It is a very strong material that is also very lightweight. Carbon fiber is five-times stronger than steel and twice as stiff. Though carbon fiber is stronger and stiffer than steel, it is lighter than steel; making it the ideal manufacturing material for many parts. These are just a few reasons why carbon fiber is favored by engineers and designers for manufacturing.

Polymer matrix composites (PMCs)

- Consists of a polymer resin as the matrix
- Being used in the diversity and largest quantities
- Glass fiber reinforced polymer composites (GFRPs)
- Produced in the largest quantities carbon fiber reinforced polymer composites (CFRPs)
- High performance composites Aramid fiber reinforced polymer composites (AFRPs)
- High strength, high modulus and high impact resistance composite.

Metal matrix composites (MMCs)

- The matrix is more ductile than reinforcements
- The reinforcement may improve specific strength, abrasion resistance, creep resistance, thermal conductivity and dimensional stability of the overall composites.

Advantage

- Higher operating temperatures
- Non-flammability
- Greater resistance to degradation by organic fluids

Disadvantage

- Expensive and higher density

Ceramic metal composites (CMCs)

- For use in high temperature and severe stress applications, e.g. automobile and aircraft gas turbine engines.

Advantage

- High strength and modulus
- Very high service temperature
- Reduced weight (lower fuel consumption)

Disadvantage

- Very brittle

Fiber reinforced composites

Fibers are the important class of reinforcements ,as they satisfy the desired conditions and transfer strength to the matrix constituent influencing and enhancing their properties as desired.

Laminar composites

Laminar composites are found in as many combinations as the number of materials. They can be described as material comprising of layers of materials bonded together. These may be of several layers of two or more metal materials occurring alternately or in a determined order more than once, and in as many numbers as required for a specific purpose.

Particulate reinforced composites

Microstructures of metal and ceramics composites ,which show particles of one phase strewn in the other,are known as particle reinforced composites. Square,triangular and round shapes of reinforcement are known,but the dimensions of all their sides are observed to be more or less equal .The size and volume concentration of the dispersed distinguishes it from dispersion hardened materials.

Properties of composite materials

- Light in weight
- Ratios of strength-to-weight and stiffness-to-weight are greater than steel or aluminium.
- Fatigue properties are better than common engineering metals.
- Composites cannot corrode like steel.
- Possible to achieve combinations of properties not attainable with metals, ceramics or polymer alone.
- Toughness is improved.
- Fabrication or production is cheaper.

Advantages

- Lower density (20 to 40%)
- Higher directional mechanical properties (specific tensile strength (ratio of material strength to density) 4 times greater than that of steel and aluminium.
- Higher Fatigue endurance .
- Higher toughness than ceramics and glasses.
- Versatility and tailoring by design.
- Easy to machine.
- Can combine other properties (damping, corrosion).
- Cost.

Application of composite materials

- In automobile industries (e.g. steel or aluminium body)
- Marine application like shafts ,hulls,spars(for racing boat)
- Aeronautical application like components of rockets,aircrafts (business,military),missiles
- Communication antenna,electronic circuit boards